

INF6953QE - Internet of Things Security

Practical Work 1 – Fall 2026

Introduction to Machine Learning for IoT Security Datasets

Instructions:

- This work is to be submitted in groups of at least 4 - maximum 5 and via Moodle only.
- The submitted report must be in PDF format. You're free to build it in any format you want, however (.docx, .odt, .tex, etc.).
- The report must contain a title page including the course title, the lab title, your names, and student ID numbers (matricule).
- The report is limited to a maximum of 7 pages. The code should be submitted as a separate file (e.g., .py or .ipynb) and will not count toward the page limit.
- The report must be submitted by the **19th of September 2026** before 23h59. A penalty of 10% will be applied for each day after that date.
- There are 4 labs in total, each worth 14 marks/7% weighting (56 marks/28% total).

Background and Introduction:

The rapid growth of IoT networks has introduced significant security challenges. IoT devices are often resource-constrained, heterogeneous, and deployed at scale, making traditional security approaches challenging to apply and leaving them vulnerable to attackers. Machine Learning (ML) offers a powerful alternative: by analyzing patterns in network traffic, ML models can detect anomalies that may signal malicious activity, often catching threats that rule-based systems miss.

This lab introduces you to the application of ML in IoT security. You will work with the IoT-23 dataset, training and evaluating several classification models such as: Decision Tree, K-Nearest Neighbors (KNN), Logistic Regression, and Linear SVM to distinguish benign from malicious network traffic. Through this process, you will gain hands-on experience with the full ML pipeline for security applications: preparing raw network data, training and tuning models, and evaluating their effectiveness at detecting real IoT-related cyberattacks.

Dataset:

This lab uses the **IoT-23 dataset**. Read the dataset description and download instructions here: <https://www.stratosphereips.org/datasets-iot23>

For this lab, choosing a scenario means selecting one scenario folder and working with its corresponding CSV file (you do not need to process the raw PCAPs directly). A companion

Scenarios Guide is provided alongside this lab, listing available scenarios with their device/malware type, duration, and file size; use it to pick a scenario that's manageable for this assignment.

1. Understanding of Dataset and Network Diagram [2 Points]

Using the accompanying Scenarios Guide, select at least five scenarios from the IoT-23 dataset. For each scenario, create a table showing:

- The type of IoT device involved (refer to the scenario's individual page on the Stratosphere IPS website for device details, e.g., smart camera, router, sensor)
- The type of attack or malware involved (e.g., DDoS, scanning, botnet)
- A brief description of the attack behavior

Also, draw a simple diagram of the network setup showing devices, traffic flow, and points where monitoring takes place.

Finally, write one paragraph explaining how IoT device traffic is monitored and the key differences between normal and malicious traffic patterns.

2. Applying ML Algorithms to IoT-23 Data [6 Points]

Data preparation:

- Handle missing or invalid values.
- Remove unnecessary columns (e.g., timestamps, non-numeric IDs).
- Encode the label as 0 = benign and 1 = attack.
- Report the class distribution (benign vs. attack) before training.
- Scale the numeric features using StandardScaler or MinMaxScaler (important for some models).
- Split the data into training (70%), validation (15%), and test (15%) sets. Use the validation set to check for overfitting; report final metrics on the test set.

Apply the following ML algorithms:

- Decision Tree
- K-Nearest Neighbors (KNN)
- Logistic Regression
- Linear SVM

Evaluate each model using:

- Confusion Matrix
- Accuracy, Precision, Recall, and F1-score

3. Comparison of Results [3 Points]

Create a comparison table showing the results for each model:

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree				
KNN				
Logistic Regression				
Linear SVM				

Write a short analysis explaining which model performed best and why. Consider factors such as accuracy, speed, interpretability, and (given the class distribution you reported earlier) whether accuracy alone is a reliable indicator of performance for this dataset.

4. Testing on a Different Scenario [3 Points]

To evaluate how well your best-performing model generalizes, train it on one IoT-23 scenario and test it on a different scenario; ideally involving a different device or attack type than the one used for training. See the Scenarios Guide for an explanation of why this matters and to help you select two distinct scenarios.

- Report the same metrics as before (Accuracy, Precision, Recall, F1-score).
- Write a short paragraph explaining how the performance changed compared to Part 3, and what this means for detecting attacks on new devices or environments.

References:

1. Zhang, Z. (2019). *Machine learning and IoT: The convergence of data and the connected world*. Wiley.
2. Yao, M. (2021). Machine learning for anomaly detection in IoT networks. *Journal of Internet of Things*, 8(2), 153–167.
3. Stratosphere IPS. IoT-23 Dataset. <https://www.stratosphereips.org/datasets-iot23>
4. Scikit-learn Documentation. <https://scikit-learn.org/stable/>

5. Scenarios Guide (companion document provided with this lab).